

INSURANCE CHARGE ANALYSIS

Key factors influencing medical insurance charges

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Aim: identify which customer factors drive insurance charges, to guide pricing and customer advice.

DATASET OVERVIEW

1,338
customer records

7
columns: 6 features + charges

0
missing values

age age of the insured person (18–64)
sex female / male
bmi Body Mass Index (weight relative to height)
children number of dependants covered (0–5)
smoker yes / no
region northeast, northwest, southeast, southwest
charges medical insurance cost billed (the target)

Business question: which factors influence charges the most, so pricing and customer advice can be based on evidence?



DATA PREPARATION

ANALYSIS APPROACH

1	**Inspect**	Loaded the CSV with Pandas; used .head(), .info(), .shape, .describe()
2	**Clean**	Checked missing values and duplicates; fixed data types
3	**Explore**	Average, median, min and max charges; groupby by smoker, sex, region, children
4	**Transform**	Created bmi_category and age_group with pd.cut()
5	**Matplotlib**	Bar (region), line (age) and pie (smokers) charts
6	**Seaborn**	Histogram, boxplot, scatter plot and correlation heatmap
7	**Insights**	Summarised the main drivers and turned them into recommendations

DATA CLEANING: RESULTS

0

missing values in any column, so no imputation needed

1

exact duplicate row found and removed

1,337

clean rows used for the analysis

3

text columns (sex, smoker, region) converted to category

The duplicate: a 19-year-old male non-smoker in the northwest (BMI 30.59, $1,639.56) appeared twice. Numeric columns were already the correct types.

NEW COLUMNS CREATED

bmi_category

Category	BMI	Count	Avg charge
Underweight	< 18.5	20	$8,852
Normal	18.5–24.9	225	$10,409
Overweight	25–29.9	386	$10,988
Obese	**30+**	**706**	**$15,572**

age_group

Group	Ages	Count	Avg charge
Young	18–34	548	$9,688
Middle-aged	35–49	404	$13,744
Senior	**50–64**	**385**	**$17,903**

Why these cut-offs: BMI uses the standard WHO bands. Age groups split the 18–64 range into three similar-sized groups, so each average is based on enough customers.

What they show: over half of customers (53%) are obese, and the average charge nearly doubles from the youngest to the oldest group.

VIEW THE FIRST 5 ROWS

	age	sex	bmi	children	smoker	region	charges
0	19	female	27.900	0	yes	southwest	16884.92400
1	18	male	33.770	1	no	southeast	1725.55230
2	28	male	33.000	3	no	southeast	4449.46200
3	33	male	22.705	0	no	northwest	21984.47061
4	32	male	28.880	0	no	northwest	3866.85520

Each row is one customer: three numeric features (age, bmi, children), three categorical features (sex, smoker, region) and the target, charges.

CHECK DATA TYPES AND NON-NULL COUNTS

#	Column	Non-Null	Count	Dtype
0	age	1338	non-null	int64
1	sex	1338	non-null	object
2	bmi	1338	non-null	float64
3	children	1338	non-null	int64
4	smoker	1338	non-null	object
5	region	1338	non-null	object
6	charges	1338	non-null	float64

What this tells us

1,338 rows × 7 columns, all fully populated.

4 numeric columns (int64 / float64) and 3 text columns (object).

Fix applied: sex, smoker and region converted to category type after cleaning.

VIEW SUMMARY STATISTICS

	age	bmi	children	charges
count	1338.000000	1338.000000	1338.000000	1338.000000
mean	39.207025	30.663397	1.094918	13270.422265
std	14.049960	6.098187	1.205493	12110.011237
min	18.000000	15.960000	0.000000	1121.873900
25%	27.000000	26.296250	0.000000	4740.287150
50%	39.000000	30.400000	1.000000	9382.033000
75%	51.000000	34.693750	2.000000	16639.912515
max	64.000000	53.130000	5.000000	63770.428010

Key point: charges are right-skewed. The mean ($13,270) is well above the median ($9,382), because a minority of very high bills pulls the average up.

CHARGES AT A GLANCE

$13,279
average charge

$9,386
median (typical) charge

$63,770
highest charge (lowest: $1,122)

1,063
non-smokers (79.5%)

274
smokers (20.5%)

3.8×
smokers' average charge vs non-smokers'

Smokers average $32,050 against $8,441 for non-smokers. They make up a fifth of customers but account for 49% of all charges.

CHARGES BY GENDER AND REGION

Gender	Customers	Avg charge	% smokers
Male	**675**	**$13,975**	**23.6%**
Female	662	$12,570	17.4%

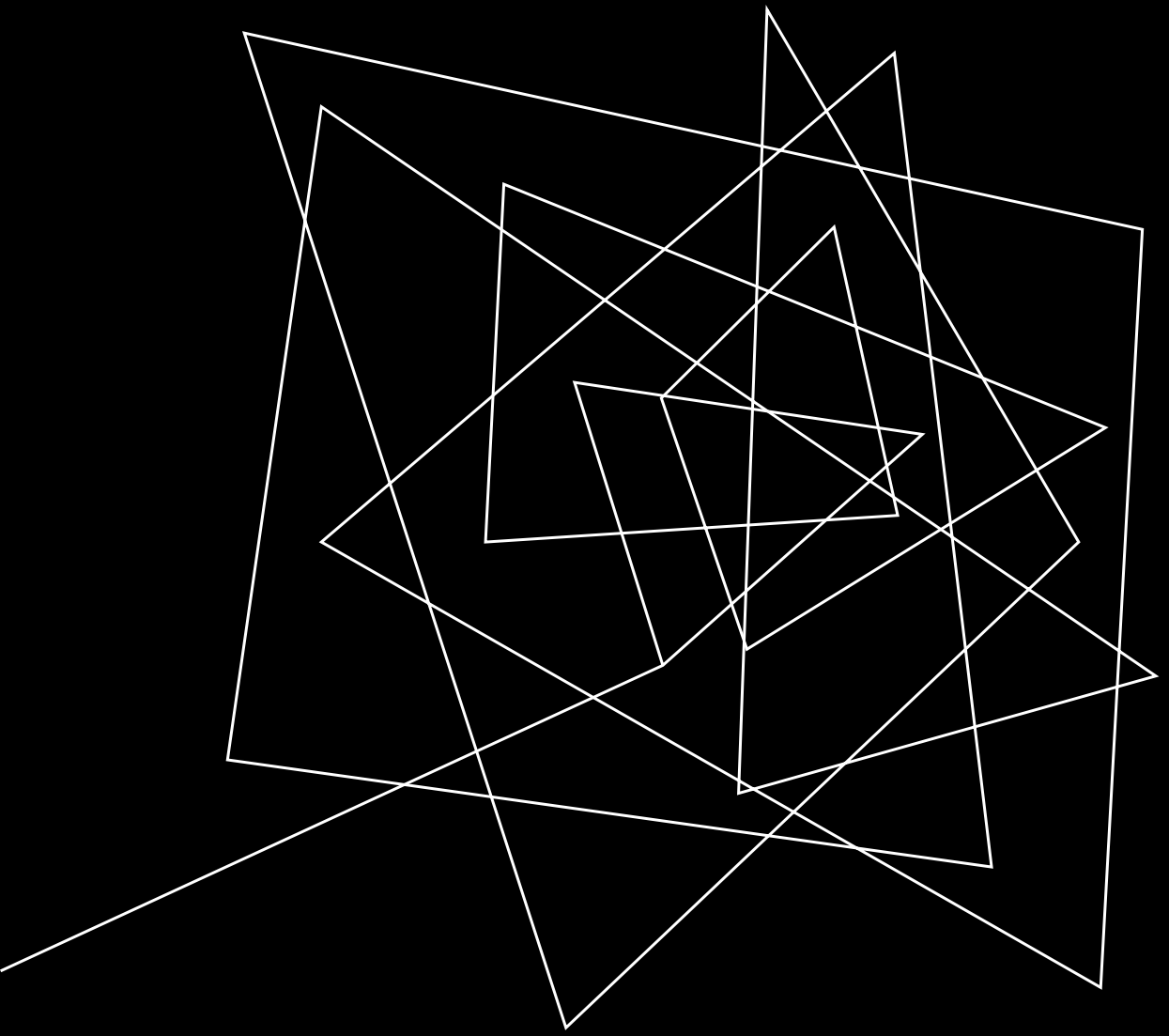
Region	Customers	Avg charge	% smokers
Southeast	**364**	**$14,735**	**25.0%**
Northeast	324	$13,406	20.7%
Northwest	324	$12,451	17.9%
Southwest	325	$12,347	17.8%

Takeaway: men and the Southeast pay more on average, but they also have the most smokers. Smoking, rather than gender or region, is the likely reason for the gap.

CHARGES BY NUMBER OF CHILDREN

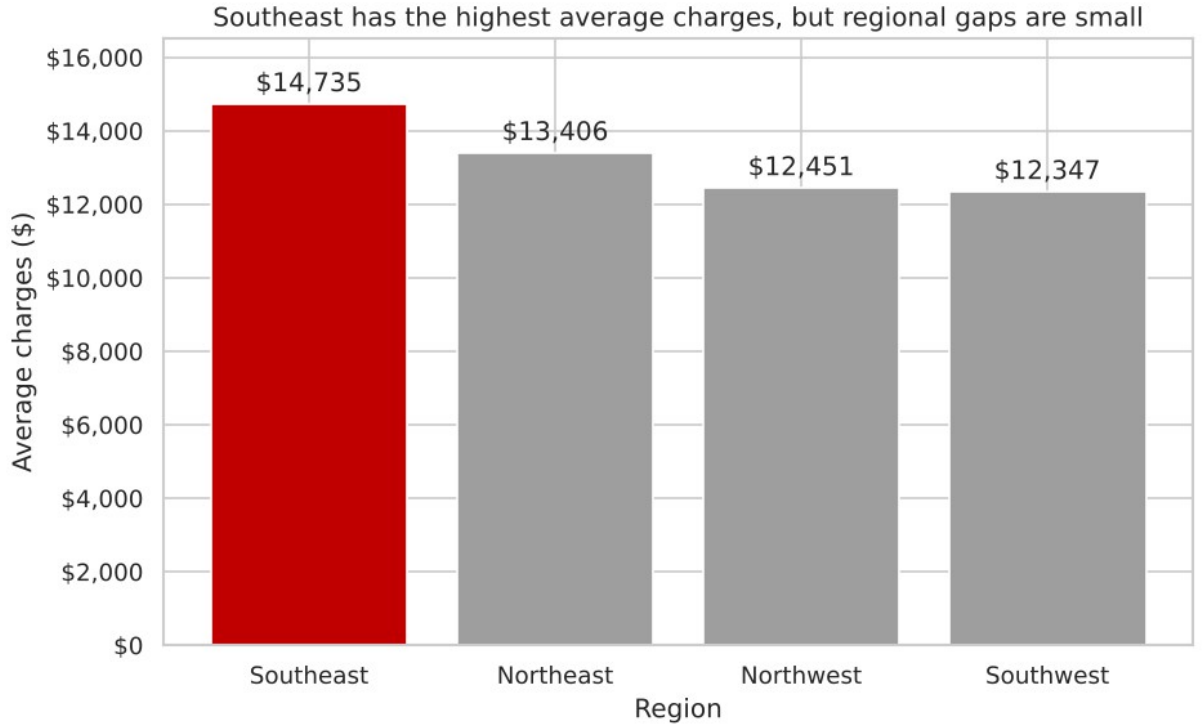
Children	Customers	Avg charge	% smokers
0	573	$12,385	20.1%
1	324	$12,731	18.8%
2	240	$15,074	22.9%
3	157	$15,355	24.8%
4	25	$13,851	12.0%
5	18	$8,786	5.6%

Takeaway: there is no clear pattern (correlation r = 0.07). The low averages for 4–5 children are based on only 25 and 18 customers, who are also mostly non-smokers, so they are not reliable.



VISUALIZATIONS

REGION HAS LITTLE EFFECT ON CHARGES



Matplotlib · bar chart

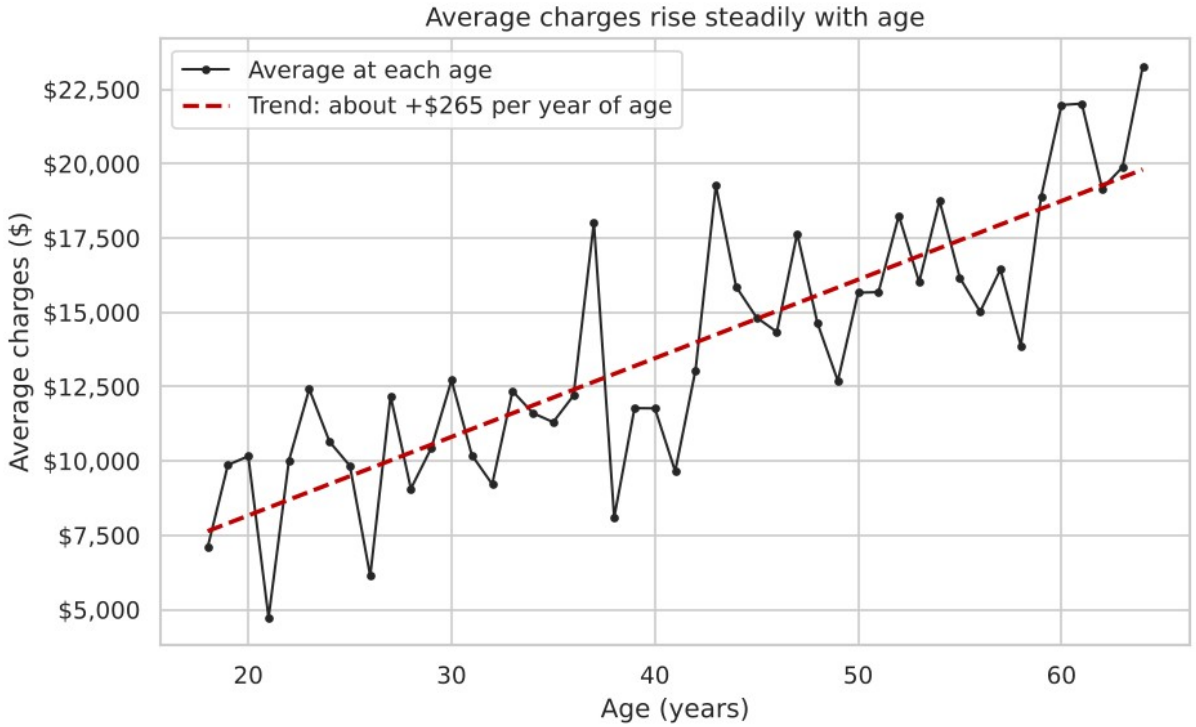
$2,388

gap between the highest and lowest region

The Southeast averages $14,735; the other regions sit between $12,300 and $13,400.

The Southeast also has the most smokers (25%) and the highest average BMI (33.4).

CHARGES RISE STEADILY WITH AGE



Matplotlib · line chart

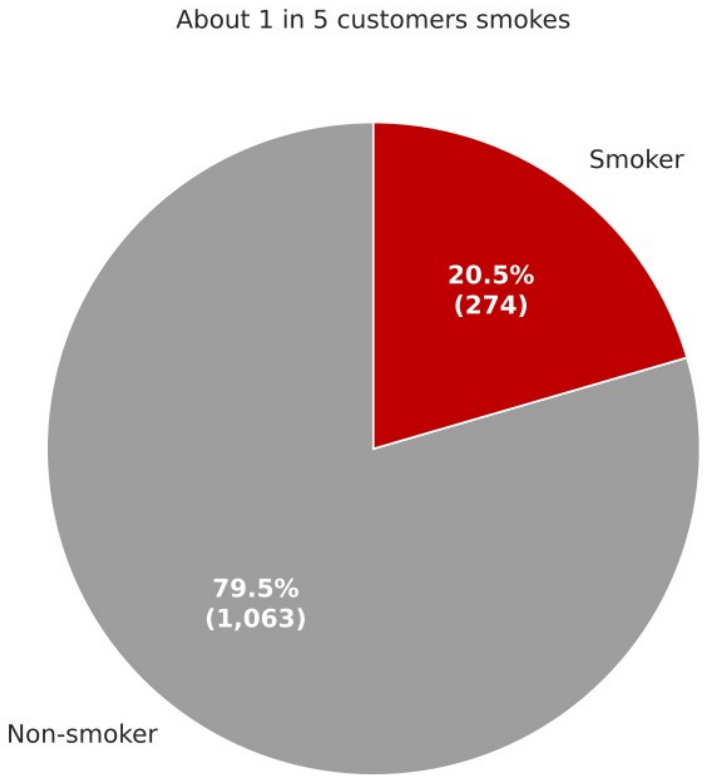
+$265

average increase per year of age

The average charge climbs from about $7,100 at 18 to $23,300 at 64.

The zig-zags come from how many smokers there are at each age.

ABOUT 1 IN 5 CUSTOMERS SMOKES



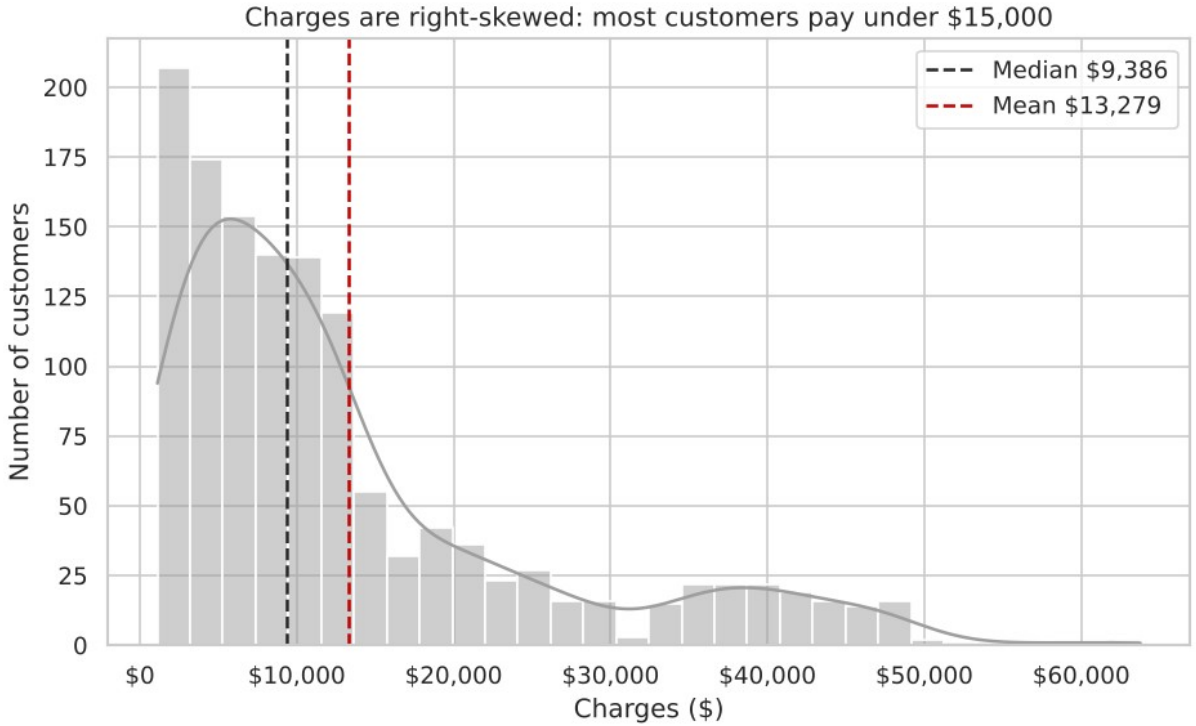
Matplotlib · pie chart

49%

of all charges come from smokers

Smokers are only 20.5% of customers (274 of 1,337), yet they generate almost half of the total amount billed.

MOST CUSTOMERS PAY UNDER $15,000



Seaborn · histogram

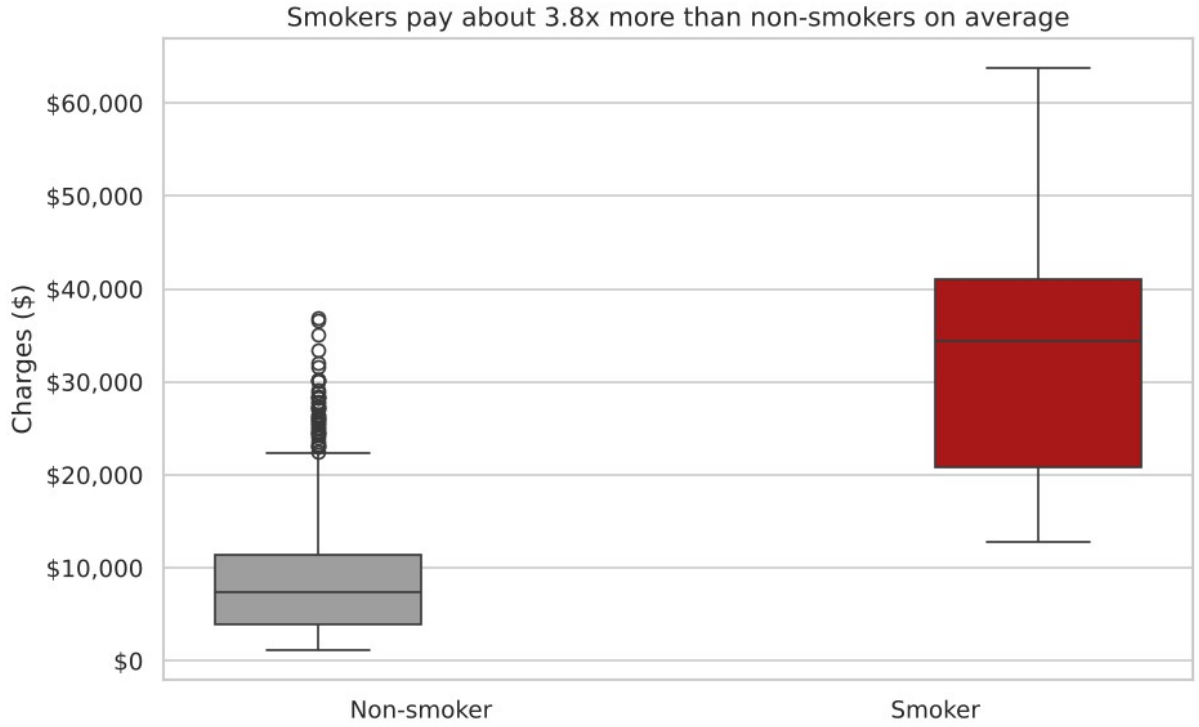
73%

of customers are billed under $15,000

Charges are right-skewed: a long tail of high bills, mostly from smokers, pulls the mean ($13,279) above the median ($9,386).

So the median is the fairer “typical” charge.

SMOKERS PAY 3.8× MORE



Seaborn · boxplot

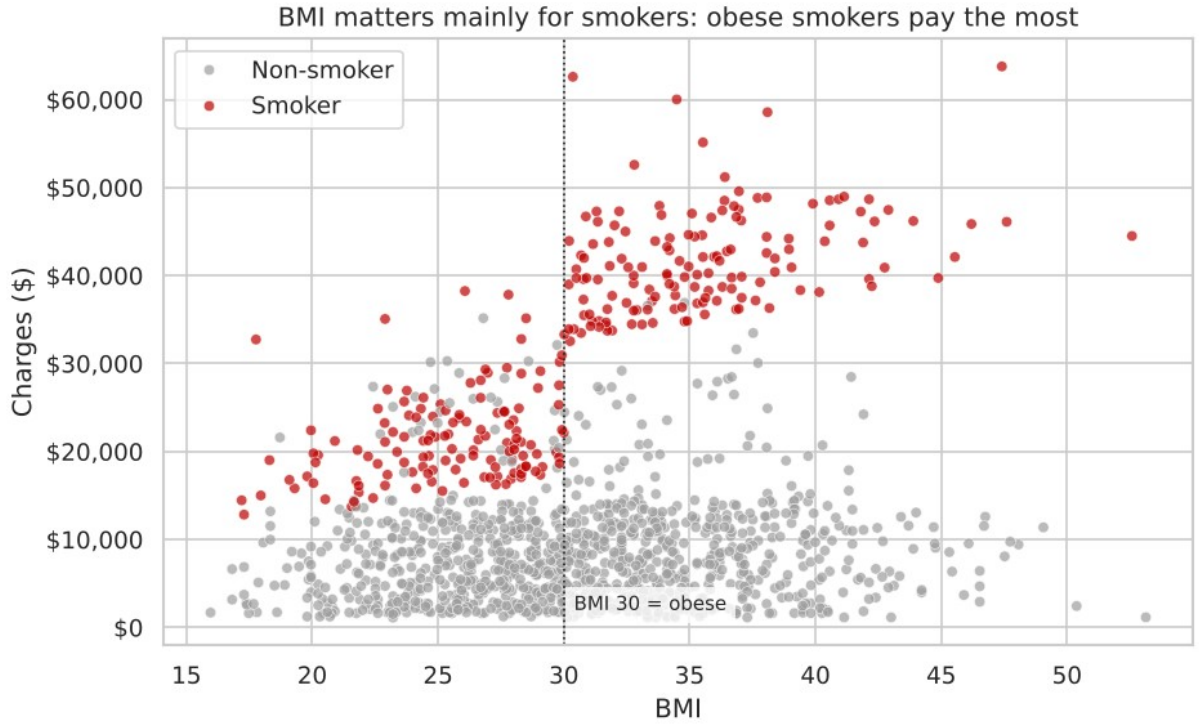
3.8×

higher average charge for smokers

$32,050 vs $8,441 on average.

Three-quarters of smokers pay over $20,800, while three-quarters of non-smokers pay under $11,400.

BMI RAISES COSTS MAINLY FOR SMOKERS



Seaborn · scatter plot

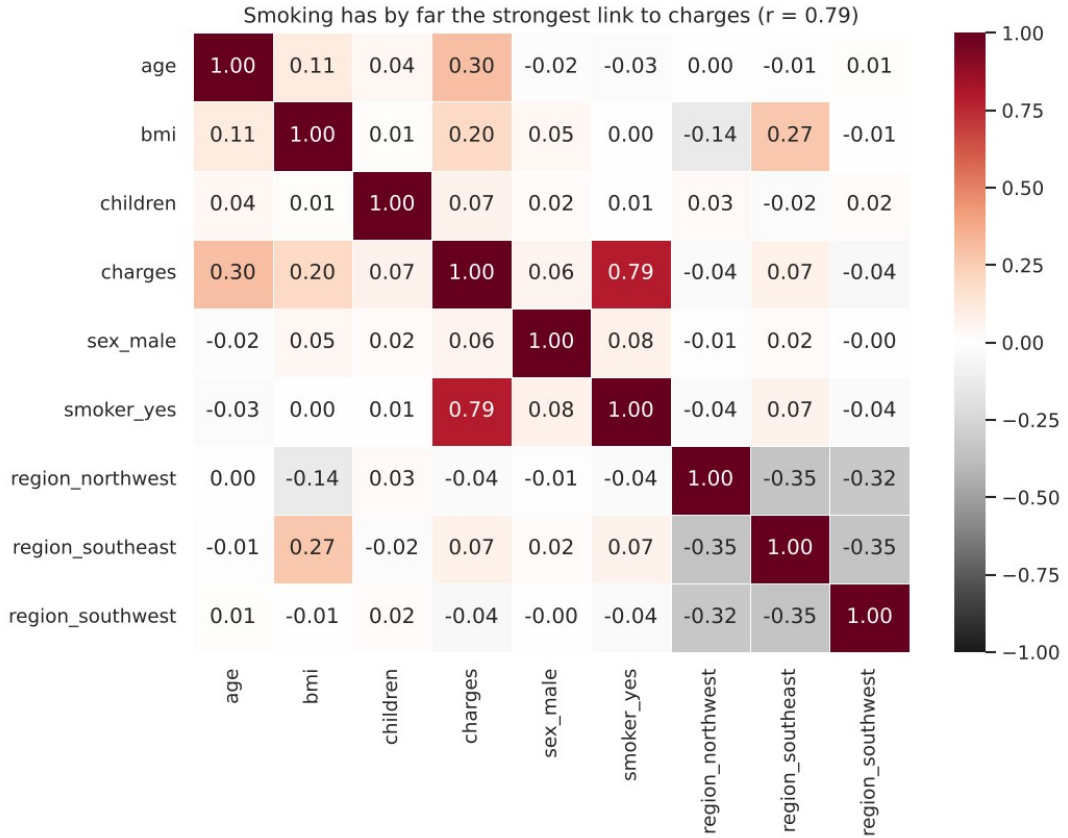
$41,558

average for smokers with BMI 30+

Nearly double the average for smokers below BMI 30 ($21,363).

For non-smokers BMI makes little difference ($7,977 vs $8,856).

SMOKING DOMINATES THE CORRELATIONS



Seaborn · correlation heatmap

r = 0.79

correlation between smoking and charges

Age 0.30 **BMI** 0.20

Children 0.07 **Sex** 0.06

Region all below 0.07

Correlation shows a link, not proof of cause.

KEY INSIGHTS

Smokers vs non-smokers

3.8×

Smokers average $32,050 vs $8,441, the largest gap in the data.

BMI effect

$41.6k

Obese smokers pay the most; BMI barely matters for non-smokers.

Age impact

+$265/yr

Average charges rise from $9.7k (18–34) to $17.9k (50–64).

Regional differences

≤ $2.4k

Small gaps that are largely explained by smoking and BMI.

BUSINESS RECOMMENDATIONS

1

Price mainly on smoking status, and verify it

Smoking is the strongest driver (r = 0.79, 3.8× higher charges). Confirm declarations with a medical questionnaire or test to prevent under-reporting.

2

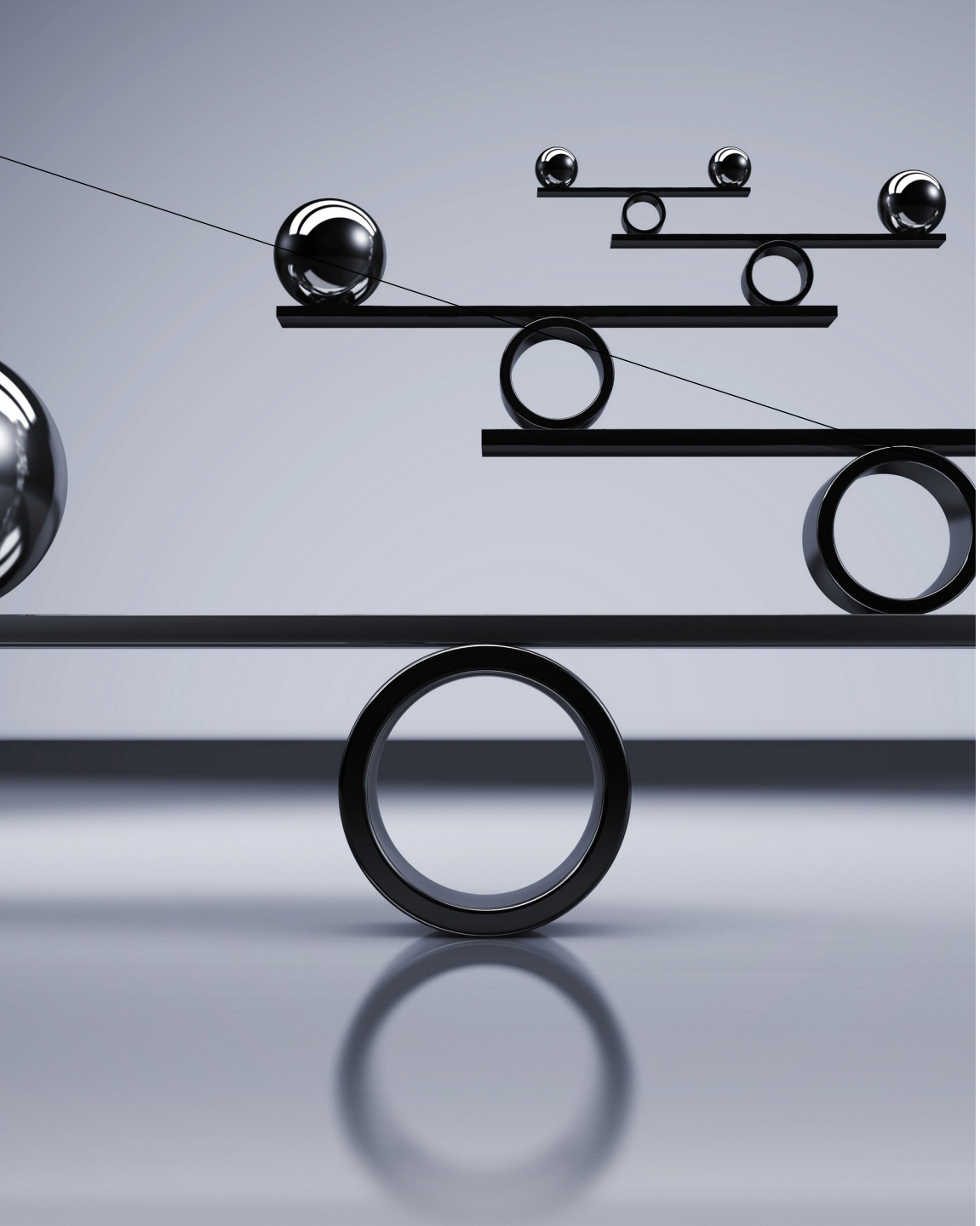
Target support at smokers with BMI 30+

This group averages $41.6k a year. Stop-smoking and weight-management programmes aimed at them could reduce claims far more than programmes aimed at everyone.

3

Use age bands; avoid pricing on region or gender

Age adds about $265 a year, a steady and predictable effect. Region and gender gaps are small and largely explained by smoking and BMI.



CLOSING & REFLECTION

Recap: cleaned 1,337 records, explored them with Pandas, built 7 charts and identified three main drivers.

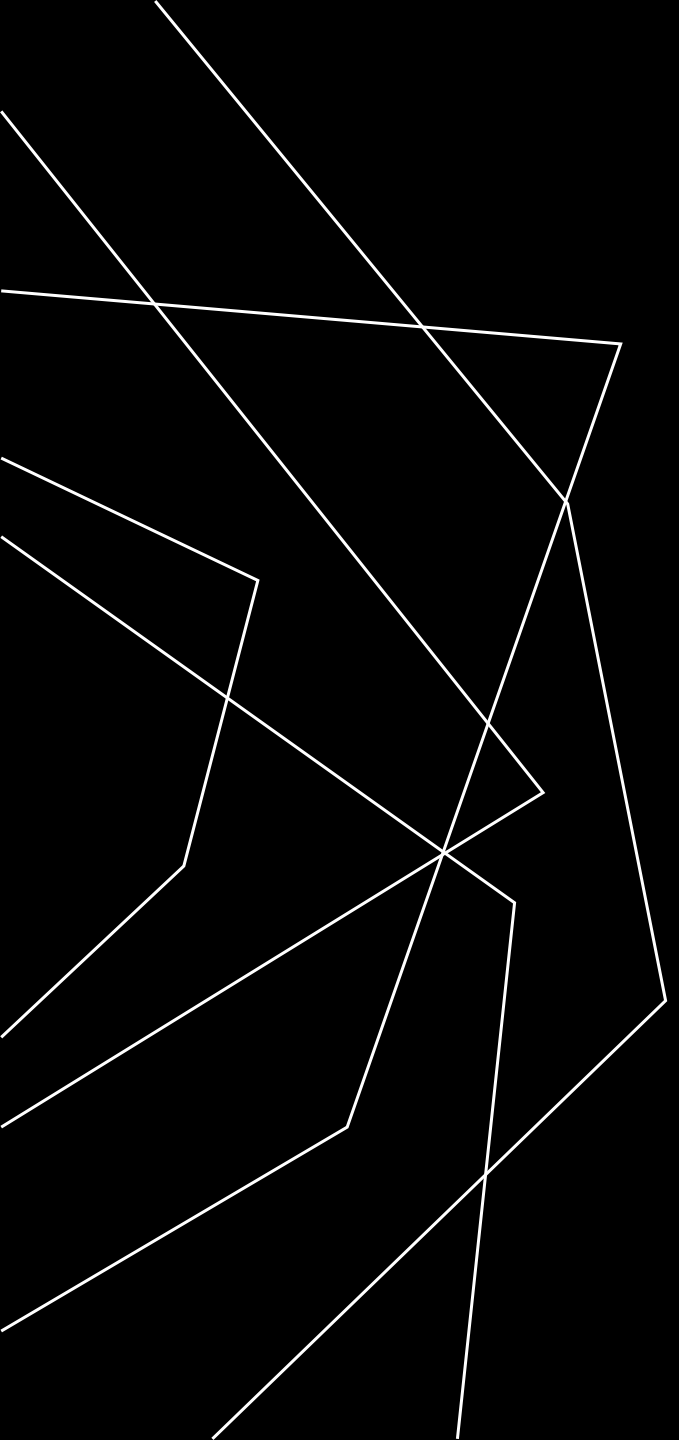
Reflection: the three strongest drivers

1. **Smoking status (r = 0.79)**
   Smokers pay 3.8× more; smoking causes costly chronic disease.

2. **Age (r = 0.30)**
   Charges rise about $265 a year as health problems become more common.

3. **BMI (r = 0.20)**
   Linked to diabetes and heart disease; strongest when combined with smoking.

Sex, region and number of children have only weak links ($|r| < 0.07$).



THANK YOU